Range Difference Between Shallow and Deep Channels of Airborne Bathymetry LiDAR With Segmented Field-of-View Receivers

Jizhe Li, Bangyi Tao, Yan He, Youzhi Li, Haiqing Huang, Zhihua Mao, and Jiayong Yu

***Abstract*— Significant range differences were identified between shallow and deep channels of the Mapper5000 bathymetry light detection and ranging (LiDAR) system with segmented field-of-view (FOV) receivers. Range difference varied with depth and water optical properties. The main feature was the maximum value in range difference curves, which ranged from 0.3 to 0.6 m and usually exceeded the International Hydrographic Organization (IHO) accuracy standards. Sensitivity analyses based on a semianalytical Monte Carlo simulation model revealed that the scattering phase function and laser beam divergence angle played more important roles in causing pulse dispersion and determining the amplitude and position of maximum range difference than absorption and scattering coefficients. A range difference correction method by fitting existing shallow and deep channel data in the overlapping range with a cubic polynomial was proposed to correct the deep channel data in the entire depth range that LiDAR can detect. Depth discontinuity at the junction of the shallow channel and deep channel measurements was successfully removed, and the mean and standard deviation of corrected range differences were within 0.01 and 0.1 m, respectively. A combination of range difference correction and mean bias corrector can be an alternative method for depth bias correction of segmented-FOV LiDAR when referenced sonar data is not available.**

***Index Terms*— Airborne LiDAR bathymetry (ALB), correction method, range difference, segmented field-of-view (FOV), semi-analytical Monte Carlo method (SALMOM).**

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I. Introduction

AIRBORNE light detection and ranging (LiDAR) bathymetry (ALB) is a proven technology for hydrographic surveying and coastal mapping [1]. This technology was conceptualized in the mid-1960s, and first-generation systems were successfully developed in the USA, Canada, and Australia in the 1970s [2]–[5]. Several fully operational ALB systems, such as Optech Scanning Hydrographic Operational Airborne Lidar Survey (SHOALS) [6] and Coastal Zone Mapping and Imaging Lidar (CZMIL) [7], [8], the Riegl VQ–820G and Riegl VQ–880G series [9], the Fugro Laser Airborne Depth Sounder (LADS) Mk series [10], Leica Hawk Eye series, and Chiroptera II [11], are in operation to date. Recently, a Mapper5000 LiDAR system for the coastal bathymetry was designed and constructed by the Shanghai Institute of Optics and Fine Mechanics. This system was initially developed in the 2000s; and the latest system update was accomplished in 2017 [12], [13].

ALB uses a scanning pulsed laser beam in the green spectrum (typically 532 nm) that can penetrate water to measure the depth of shallow coastal areas from a low-altitude aircraft. One of the biggest problems that must be solved in the design of a bathymetric LiDAR receiver is the handling of more than six orders of magnitude of amplitude dynamic range between strong surface returns and weak bottom returns [14]. Using a segmented field-of-view (FOV) receiver in the ALB system is a prevailing method to solve this challenge and improve the maximum depth penetration capability [15], [16]. The CZMIL design has a large annular FOV (6–40 mrad) receiver (deep-channel) directed at deep-water signals and a center FOV (6 mrad) receiver (shallow-channel) mostly significant in shallow-water conditions. The center receiver channel is further divided into seven individual segments (1.9 mrad) [17]. This segmented FOV design can successfully handle the large dynamic range to make a seamless transition between measurements in the seafloor and terrestrial region, which makes CZMIL capable of generating high-resolution maps in shallow water without compromising seafloor detection in deeper waters [8], [16].

The segmented FOV method takes advantage of propagation-induced stretching properties in the scattering medium of the ocean [18]. The incident laser beam is scattered in the water column and spatially spread out into an expanding cone, and its size exponentially increases with increasing optical depth. Accordingly, the laser beam

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energy is mostly distributed in the shallow channel when the physical depth or optical thickness is minimal, and more energy is spread into a deep channel with a larger annular FOV with increasing depth. The deep-channel photomultiplier tube (PMT) can avoid saturation effects caused by the strong surface return because of this spatial spreading feature and apply a high gain to detect the very weak bottom return signals at large depth [14]. However, the temporal dispersion of laser pulse, together with spatial spreading, can cause depth/range measurement biases [18]. To avoid confusion with the definition of "range difference" between channels, "depth bias" in this study refers to the deviation of the LiDAR depth retrieval from the "true" value, such as sonar depth. The occurrence of depth bias is caused by the lengthening of the total integrated path length of the laser pulse that traverses the water column twice due to the multiscattering transport mechanism [19], [20]. The LiDAR range measurement in the water is determined by the round-trip transit time interval between the peaks of the sea surface and bottom reflections. The peak time is the basic information used to retrieve the range or depth in most signal processing and pulse location algorithms [21]–[25]. The lengthening of the total integrated path length results in a shift of the peak time, inducing biased in-depth estimates [17], [26]. The depth biases are not only associated with optical properties, such as optical depth, scattering phase function, and single-scattering albedo, but also affected by instrument parameters, such as receiver FOVs. Therefore, the biases in channels with various receiver FOVs differ; in other words, there is a range difference (equivalent to bias difference) between shallow and deep channels.

Currently, only a few articles covered this topic of range difference between channels [14], [17], [27]. Guenther *et al.* [14] suggested that the range difference between shallow and deep channels in the SHOALS system was owing to different FOVs and alignment, but the deep channel in the SHOALS is a full FOV channel rather than a segmented one. Ramnath *et al.* [17] indicated the viewing geometry of the CZMIL segmented receiver introduces propagation differences between channels and concluded that these differences are systematic and can be determined in the laboratory and removed before the selection of the optimum water surface detection time is made. This means that the correction is only carried out for the range difference in the air, not for that in water. In fact, the round-trip path that the laser pulse travels to the bottom and back varies greatly in different channels [18], and the difference in water cannot be predicted in the laboratory. Currently, the calibration experiment for shallow and deep channels of the new Experimental Advanced Airborne Research Lidar (EAARL-B) [28] was implemented in clear water with a diffuse attenuation coefficient (K_d) near 0.1 m^{-1} by using the sonar data [27]. Although a good agreement between its deep and topobathy (shallow) channels was found over the range of 1–16 m, their difference at a greater depth of 16–34 m was not revealed since its topobathy channel was only optimized for shallow water, and Wright *et al.* [27] also indicated that the results may be different for other water bodies with a higher K_d value. More importantly, the deep channel in the Experimental Advanced Airborne Research Lidar (EAARL)-B is still the one with a full FOV of up to 50 mrad, encompassing all its three laser-illuminated spots. Thus, the range difference between channels with segmented FOV receivers remains unclear.

The most accepted procedure is using sonar depth data as a reference to correct the range difference between channels and correct depth bias for each channel simultaneously [27], [29]. However, sonar depth data are not always available at all survey sites. It is necessary to propose another more practical method to correct depth biases in shallow and deep channels of segmented-FOV bathymetry LiDAR. The shallow-channel receiver primarily captures scattered signals from the reference path, which is the unscattered ray in the water medium [18]; thus, the depths from a shallow channel are probably closer to the "true" depths than those from a deep channel with the annular FOV. The compromised solution for depth bias correction of segmented FOV LiDAR is first to minimize the range difference of the deep channel from the shallow channel and then apply the mean bias corrector function proposed by Guenther and Thomas [18] to correct depth bias in the shallow-channel data and deep-channel data after the range difference correction.

The Mapper5000 LiDAR system adopts a similar optical design of segmented FOV receivers as the CZMIL system and thus similarly includes a shallow channel with a small FOV of 6 mrad and a deep channel with a ring form FOV of 6–40 mrad. Recently, Mapper5000 has been frequently used to carry out flight experiments in water bodies with various optical properties. Significant differences in range estimates in water between its two channels were identified in the experimental results, and thus obvious depth discontinuity at the junction of the two channel measurements was also found. The range difference between channels showed large variation with depth and had very complicated behaviors under different environmental conditions, which exceed the accuracy standards of the International Hydrographic Organization (IHO). However, whether these complicated range difference trends are caused by the instrument faults or by the radiative transfer process of laser in water is still unclear. Furthermore, due to the small FOV and limited maximum depth detection ability of the shallow channel, range differences between channels at greater depths without shallow-channel data, are not available. Thus, estimation of range difference between channels in the entire depth range that Mapper5000 detect is the greatest challenge in the procedure of depth bias correction.

In this study, to delineate the relationship of range difference to depth, we first comprehensively analyzed field measurements from several regions with various optical properties. Second, we relied on simulation results to prove that the observed relationships are reasonable from the perspective of the radiation transfer mechanism. On top of that, simulation results can further reveal the changing tendency of range differences at larger depths. To obtain simulation results under different environments, a more flexible Monte Carlo model, which is similar to the one proposed by Eren *et al.* [30], was developed to acquire more realistic simulations for the Mapper5000 LiDAR system with oblique incidence and thus the leading effect of beam spreading in both spatial and temporal dimensions on the physical causes of these range difference

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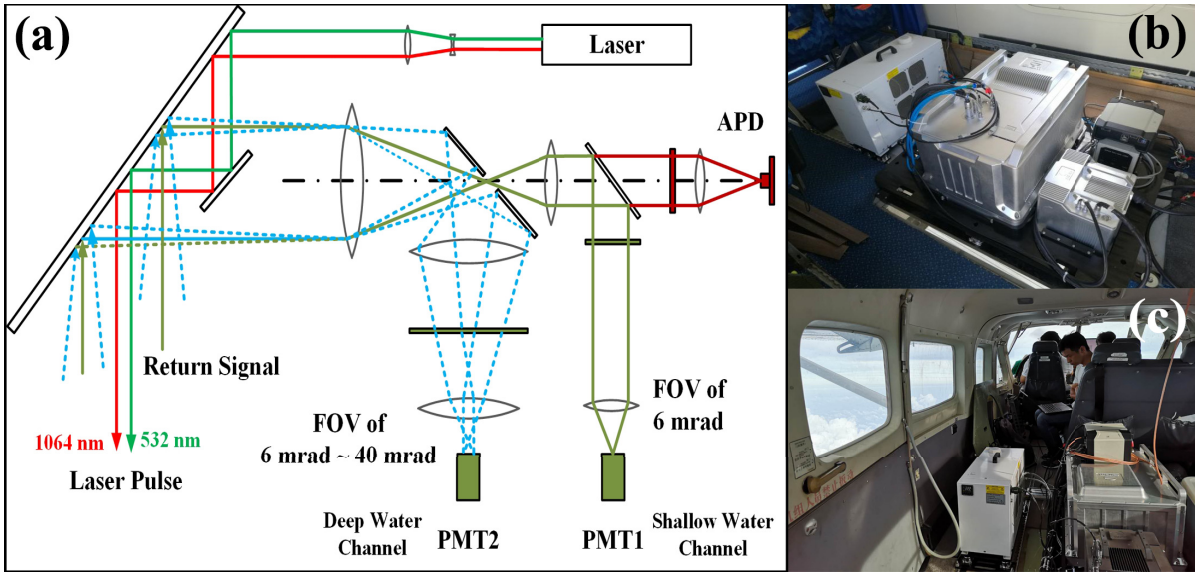

Fig. 1. (a) Schematic sketch of the entire optical system of Mapper5000. (b) Entire Mapper5000 system, including the positioning and navigation units. (c) View of Mapper5000 inside the aircraft.

could be addressed since current beam spread function for a pulsed laser beam with time dispersion [20], [31], [32] generally invokes the small-angle scattering approximation and has limits in characterizing the time dispersion for off-nadir LiDAR system. Finally, we presented a model for range difference correction for the deep-channel data in the entire depth range and access the performance of the mean bias corrector.

This article is organized as follows. In Section II, we describe the experimental dataset and processing method of Mapper5000, including detail of the Monte Carlo model for LiDAR. In Section III, we present range difference results from field measurements and fully discuss the range difference in the entire depth range under various environmental conditions based on model simulations. We also propose a correction method for the deviation of deep-channel depth from shallow-channel depth. In Section IV, we conclude and discuss required future improvements on the depth bias correction for shallow-channel depth.

II. Data and Methods

A. Mapper5000 LiDAR System

Similar to other general airborne bathymetric LiDAR systems [7], [9], [14], [33], [34], Mapper5000 (Fig. 1) is equipped with an Nd:YAG laser with an infrared output of 1 mJ at 1064 nm and a simultaneous, coaxial, frequency-doubled bluish-green output of 3 mJ at 532 nm. The pulse repetition rate is 5 kHz, and the pulsewidth is 0.95 ns. The beam divergence angle (DIV) expands to 0.3 mrad. Mapper5000 uses segmented FOV receivers, the optical design of which was first introduced in the CZMIL system, to achieve the ability to detect optical signals over a high dynamic range [7], [8], [35], [36]. The layout of the optical system of Mapper5000 [Fig. 1(a)] can be regarded as a monostatic LiDAR, in which the optical axes of the laser and receiver coincide. The return signals are focused by a lens onto a field separator. The field separator divides a full FOV (40 mrad) into two parts of a small FOV (6 mrad) at the center and a ring form FOV (6–40 mrad). The small FOV, known as a shallow channel, applies low gain to prevent surface return saturation for the detection of shallow depths and determination of topographic height. Meanwhile, the ring form FOV, the so-called deep channel, applies high gain for the weak bottom return detection at a larger depth. Then, two PMT detectors and avalanche photodiodes (APDs) are employed in its optical receiver module to detect optical signals for each channel. The scanner is a nutating design similar to SHOALS [14], whose mirror axis is approximately 7.5° offset from the axis of rotation. The resulting pattern on the earth's surface is a tightly interlocked series of pseudo-ellipse. The positioning and navigation unit consists of a kinematic GPS (KGPS) and a NovAtel POS 100C. The other system parameters of Mapper5000 are listed in Table I.

B. Experimental Data

Three flight experiments (Fig. 2) were conducted on Ganquan Island, Dazhou Island, and the Weihai coast, respectively. These sites cover different water types with various optical properties. The LiDAR attenuation coefficients (K_{LiDAR}) for the three sites are 0.077, 0.19, and 0.39 m^{-1}, respectively. The first site was Ganquan Island [Fig. 2(a)], which is located in the southeast of Hainan Province (111°35′ 10″E, 16°30′ 28″N), and its bottom is mainly formed by high reflective coral reefs. The LiDAR data were acquired over Ganquan Island from 12:30 to 18:00 on August 28, 2020, with an unmanned aerial vehicle platform. The flight height was 500 m to maximize its area coverage. The second site [Fig. 2(b)] was Dazhou Island, which is in the southeast of Wanning (110°23′ 28″E, 18°47′ 43″N), Hainan Province. The flight mission parameters did not change. The experiment was conducted from 9:00 to 12:00 on October 2, 2021.

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TABLE I
SYSTEM PARAMETERS OF MAPPER5000

Laser type	Nd:YAG infrared: 1 mJ at 1064 nm Blue-green: 3 mJ at 532 nm
Laser pulse repetition rate	5 kHz
Laser pulse width (full width at half maxima of laser)	0.95 ns (dominated by eye-safety limitations and surface return optimization)
Scan type	Off-axis scanner mirror with a scan angle from 10.5° to 15°
Scan rate	600 RPM (variable)
Scan pattern	Pseudo-ellipse
Laser receiver	Telescope with a large receiver aperture (diameter of 300 mm)
1 APD detector	1064 nm
2 PMT detector	532 nm (shallow and deep channels)
Waveform digitization rate	1 GHz

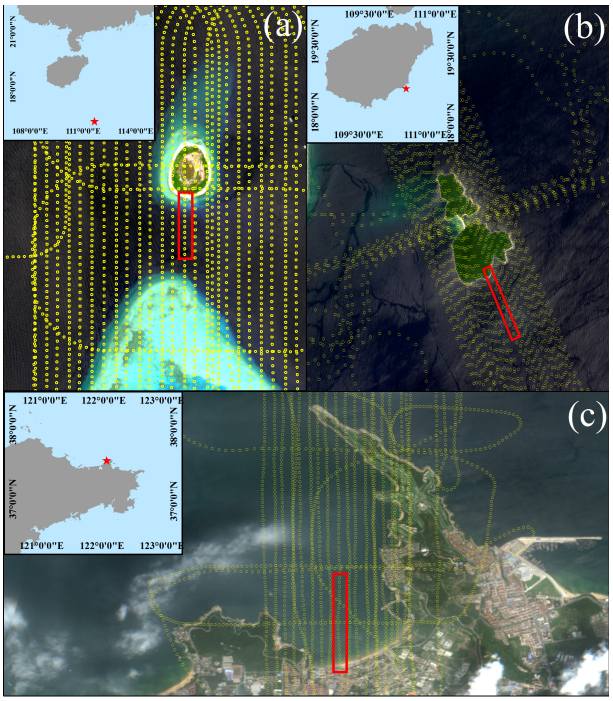

Fig. 2. (a) Three experimental sites. (a) Ganquan Island in the South China Sea. (b) Dazhou Island in Hainan Province. (c) Weihai coast in Shandong Province. The flight trajectories are shown in yellow circles, and the study areas are denoted by red rectangles.

The third site [Fig. 2(c)] was located near the coast of Weihai, Shandong Province (122°3′ 48″E, 37°33′ 6″N). Mapper5000 was installed on an Isolair Helicopter, and the flight time was from 13:00 to 15:30 on August 30, 2020. Shipborne sonar depth measurements were simultaneously conducted on the Weihai coast and Dazhou Island. The data of the three experiments are presented in Table II. At each site, the experimental area was selected in a confluence of ocean and land, with a depth ranging from shallow to deep (red rectangles in Fig. 2). Additionally, several sample waveforms of CZMIL were obtained from the China Aero Geophysical Survey and Remote Sensing Center for Natural Resources for comparison with those of Mapper5000, which were measured by the CZMIL Nova II system [37].

C. Waveform Process Method

The range in the water is determined by the time interval between the peaks of the sea surface and bottom reflections. Parrish *et al.* [38] compared range determination performance for three types of full-waveform algorithms, Gaussian decomposition (GD) [39]–[41], expectation-maximization (EM) deconvolution [23], [24], [42]–[44], and a hybrid deconvolution GD approach [38]. The results indicated that there is no single best waveform processing strategy across all applications, but they found that if the system and noise parameters are known and stable, the EM-based deconvolution algorithm provides the best performance. Therefore, in this study, an exponentially modified Gaussian (EMG) method proposed by Wong and Antoniou [42] was employed for peak detection and range determination. The laser beam propagation process in the water body followed the exponential attenuation law. The trailing edge was distorted by the volume backscattering and the total integrated water attenuation of the propagation path. Accordingly, the waveform deviated to the right. The EMG function agrees well with this type of asymmetrical waveform. The EMG function is obtained by the convolution of a standard Gaussian function and an exponential function [42], given by

$$P_{\mathrm{EMG}}(t) = f_1(t) * f_2(t) \tag{1}$$

where f_1 is the standard Gaussian function

$$f_1 = A_G \exp\left[-(t-\mu_G)^2/2\sigma_G^2\right] \tag{2}$$

where A_G is the maximum amplitude of the Gaussian function, μ_G is the corresponding time, and σ_G is the standard deviation. f_2 is the exponential function expressed by

$$f_2 = \frac{1}{\tau}\exp\left(-\frac{t}{\tau}\right), \quad (t > 0) \tag{3}$$

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TABLE II
CHARACTERISTICS OF ALB EXPERIMENT DATASETS

Site	Time	Location	Flight height (m)	K_{LiDAR} (m^{-1})	Point density (bathymetry; pts/m^2)
Ganquan Island	12:30–18:00 28th August 2020	111°35′10″E, 16°30′28″N	500	0.077	0.9
Dazhou Island	9:00–12:30 2nd October 2021	110°23′28″E, 18°47′43″N	400	0.19	1.2
Weihai coast	13:00–15:30 30th August 2020	122°3′48″E, 37°33′6″N	300	0.39	1.5

where τ is determined by water optical properties. The convolution result is shown as follows:

$$P_{\mathrm{EMG}}(t) = \frac{A_G}{\tau}\int_0^{\infty} e^{-t'/\tau}\exp\left\{-\left[(t-\mu_G)-t'\right]^2/2\sigma_G^2\right\}dt'. \tag{4}$$

Given that the bottom returns are also affected by multiscattering, the surface and bottom returns were all fitted by the exponentially modified Gaussian (EMG) function. In the shallow channel, the Mapper5000 waveforms represented by the function $P_T(t)$ are formed by overlapping the EMG curves of the sea surface and bottom, namely

$$P_T(t) = P_{\mathrm{EMG,surface}}(t) + P_{\mathrm{EMG,bottom}}(t). \tag{5}$$

To minimize the error between the measured waveform and the fitting curve and to obtain the best parameter estimation, the trust-region algorithm [45] was used for the decomposition of each Mapper5000 into the surface and bottom EMG curves.

In deep channel waveform processing, only the bottom return was fitted by the EMG function to determine the bottom position. Meanwhile, the surface position of the deep channel was the same as that of the corresponding shallow channel. A systematic range difference in the air between the two channels is within ±0.064 m, which is one order of magnitude smaller than that in the ocean, due to the good optical design and alignment. Accordingly, the surface position of the shallow channel was also regarded as the same as the deep channel's. We take the local peak positions of the two waveforms as the location of the surface or bottom to calculate the range in the ocean. Then, we calculated the range difference between the two channels.

D. Semianalytical Monte Carlo Method

A semianalytical Monte Carlo method (SALMOM) was employed in this study to assess the characteristics of range differences under various environments. This method has been developed and verified [46], [47]. In contrast to the ordinary Monte Carlo method, an approach called the "method of statistic estimation" is used in the SALMOM by combining the stochastic and analytical techniques, resulting in an effective LiDAR system modeling [30], [46]. The expected value of the fraction of photons backscattered and captured by receivers is given as follows:

$$E = \frac{p(\theta')}{4\pi}\Delta\Omega\exp(-cL)T_s \tag{6}$$

where $p(\theta')$ is the volume scattering function (VSF), which is assumed to be constant over the small solid angle $\Delta\Omega$. This factor indicates the portion of photons scattered back into $\Delta\Omega$ at θ'. Variable c is the beam attenuation coefficient of the water column, and $\exp(-cL)$ is the probability that photons are directly transmitted to the air–water interface at point S without further interaction with the water column. T_s is the Fresnel transmittance of the air–water interface [46].

The geometric model of the oblique and monostatic LiDAR system was employed in the SALMOM to achieve more realistic simulations for the Mapper5000 [Fig. 3(a)]. Some adjustments had been made in the SALMOM. First, the scanner nadir angle was set to 15°, and the projection of receiver FOV on the sea surface was changed from circle to egg shape [Fig. 3(c)]. The egg shape is presented by the equation below

$$f(x,y,z) = \frac{(x-x_c(z))^2}{(e_a-e_b)(x-x_c)+e_a\cdot e_b} + \frac{(y-y_c)^2}{e_c^2} \tag{7}$$

where e_a, e_b, and e_c are the three axial lengths of the egg shape defining the boundary of the receiver FOV at a certain depth; z. x_c, and y_c are the central point coordinate of the egg shape; and (x, y) is the location where the photon is scattered or reflected at a certain depth z. Then, the sign function was used to determine whether the photon backscattered was in the receiver FOV, which can be presented as follows:

$$\mathrm{sign}(f(x,y,z)) = \begin{cases} +, & f \le 1 \\ -, & f > 1. \end{cases} \tag{8}$$

When the sign function was positive, the photon was in the receiver FOV; otherwise, it was outside the receiver FOV (not detected by the receiver).

Second, the Gaussian distribution instead of point source was employed in modeling laser pulse emission, and the circ function was adopted as the characteristic of the LiDAR receiver. The Gaussian distribution emitting laser pulse is presented as follows:

$$I_E = \frac{4}{\pi\,\mathrm{DIV}^2}\exp\left(-\frac{\left|\vec{n}-\vec{n}_{15}\right|}{\mathrm{DIV}^2}\right) \tag{9}$$

where DIV is the laser beam divergence angle, and n_{15} is the vector of the 15° emitting trajectory. The circ function for the

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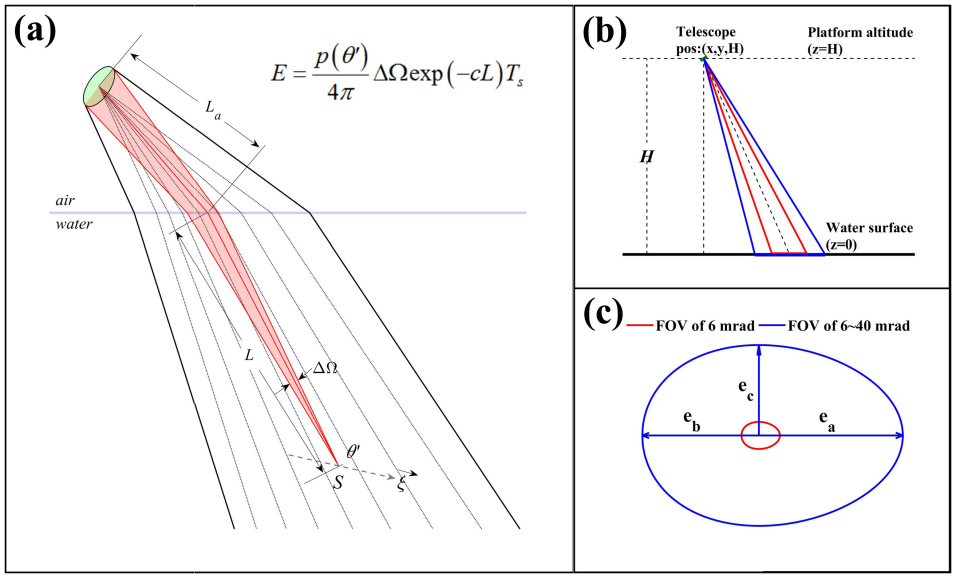

Fig. 3. (a) Schematic sketch of the SALMOM for the oblique and monostatic LiDAR system. (b) Illustration of two FOVs for shallow and deep channels. (c) Projection of oval receivers FOV.

TABLE III
CHARACTERISTICS OF ALB EXPERIMENT DATASETS

FOV	6 and 40 mrad
Incident angle	15°
Flight height	300 m
Photons number	1E8~1E10
Radius of the detector aperture	0.3 m
Sea surface	Flat (at z = 0)
Absorption coefficient	Variable
Scattering coefficient	Variable
Asymmetric parameter of VSF (g)	0.80–0.95
Seabed	Lambertian reflector
Reflectance of the seabed	0.3

LiDAR receiver is presented as follows:

$$I_R = \frac{4}{\pi \cdot \mathrm{FOV}^2}\chi\left(\left|\vec{n} - \vec{n}_{15}\right| - \frac{\mathrm{FOV}}{2}\right) \tag{10}$$

where χ is illustrated as follows:

$$\chi(x) = \begin{cases} 1, & x \le 0 \\ 0, & x > 0. \end{cases} \tag{11}$$

Fig. 3(b) and (c) illustrates the FOVs of the shallow and deep channel receivers on the surface. The inherent optical properties of the water body were assumed to be vertically homogeneous, containing the attenuation coefficient (c), single scattering albedo (ω_0), and scattering phase function. The Henyey–Greenstein (HG) phase function was applied, in which the asymmetry factor g was used to imply the energy distribution in the scattering process. The effect of the atmosphere was neglected in this model. The seabed strictly followed the Lambert law of reflection. Table III lists detailed simulation parameters.

The optical refraction path wherein the laser radiation traverses the air–sea surface was carefully considered [Fig. 3(a)], to account for off-nadir geometric stretching and off-axis path shortening or lengthening [18]. The challenge is the analytical calculation of the shortest upwelling propagation path (L and L_a) from point S to the receiver. To simplify this calculation, a number of candidate upwelling trajectories were precalculated to generate a lookup table for use in the simulation [dotted lines in Fig. 3(a)]. When a scattering or reflection event occurred at point S inside the FOV of the receiver, the candidate upwelling trajectory closest to S will be selected to determine the upwelling propagation path length and acceptable solid angle in (6).

The needed impulse responses for different FOVs under various environments were generated based on the SALMOM. Then, the simulated impulse responses were convolved with a realistic outgoing laser pulse to yield the expected waveform (P). The waveform for the deep channel (P_{6-40}) with the ring form FOV (6–40 mrad) was obtained by subtracting the waveform (P_6) for the shallow channel with 6 mrad FOV from the one (P_{40}) with 40 mrad FOV, which is given by

$$P_{6-40} = P_{40} - P_6. \tag{12}$$

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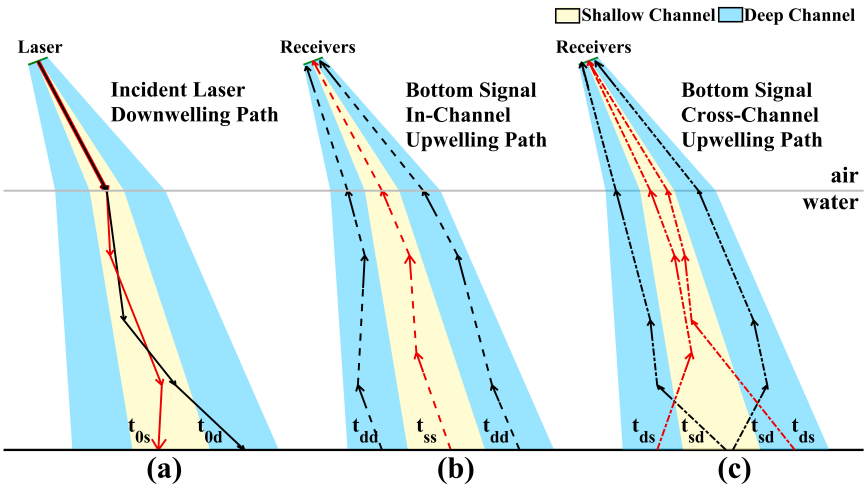

Fig. 4. Multiple scattering propagation paths of the LiDAR signal and their average multipath delay times. (a) Downwelling path from the laser to bottom. (b) In-channel upwelling path from the bottom to the receiver. (c) Cross-channel upwelling path from the bottom to the receiver.

The total round-trip propagation for the laser signal consisted of the downwelling propagation path from the laser to the bottom and the upwelling path from the bottom to the receiver (Fig. 4). In the segmented FOV channels, the upwelling path can be further divided into two parts, namely in-channel and cross-channel paths [Fig. 4(b)–(c)]. Either downwelling or upwelling propagation path time was closely related to the beam divergence and FOVs. Only the mean arriving times (MATs) from the laser to the bottom and from the bottom to the receiver were calculated by using the Drop-Dead forward Monte Carlo model for simplification [30], [48]. The downwelling MAT can be easily calculated; however, the principle of "reciprocity" [49] was involved in the upwelling MAT. Reciprocity is a statement of symmetry or reversibility that implies the ensemble of viable scattering paths in the water is identical for downwelling and upwelling radiation; this notion means that the exiting photons must leave the medium in the opposite direction from which they entered to reach the distant receiver collocated with the laser source [18]. The circ function, which was designed for the LiDAR receiver, was used instead of the Gaussian distribution to model the laser source during the upwelling propagation simulation.

The range difference arises from different integrated lengths of each propagation path. The average propagation time of these paths will directly reflect the pulse stretching effect on the increasing path length. Accordingly, an average multipath time delay (AMTD), basically defined as the difference between the MAT of a bunch of photons and the time of unscattered photons arriving at the bottom or receiver [31], was calculated to evaluate the contribution of different propagation paths to the range difference. In Fig. 4, the pale yellow and pale blue blocks represent the received FOVs of the shallow and deep channels, respectively. The red and black lines represent the trajectories of multiple scattering photons associated with shallow and deep channels, respectively. The incident photons emitted by the laser device are also presented. t_{0s} and t_{0d} in Fig. 4(a) indicate the AMTD of the laser pulse arriving at the bottom in the FOVs of the shallow and deep channels, respectively. The "In-Channel" upwelling AMTD [Fig. 4(b)] implies the time for the photons to reflect at the bottom in a certain channel FOV transited into the same channel receiver. Meanwhile, t_{ss} and t_{dd} represent the In-Channel upwelling AMTD shallow and deep channels, respectively. Conversely, the "Cross-Channel" upwelling AMTD in Fig. 4(c) indicates that the photons reflected at the bottom in a certain channel FOV transited into another channel receiver. t_{sd} represents the Cross-Channel upwelling AMTD of the photons reflected at the bottom region in the shallow-channel FOV and captured by the deep-channel received FOV, while t_{ds} is the opposite. The principle of "reciprocity" was adopted to model these upwelling propagation AMTD, as mentioned in Section II-D. The total AMTD for the shallow channel (t_s) and deep channel (t_d) was proportional to the summation of AMTD for the above-mentioned three paths with different weights, which can be roughly represented by

$$\begin{aligned} t_s &\propto W_s t_{0s} + W_s \cdot t_{ss} + W_d \cdot t_{ds} \\ t_d &\propto W_d t_{0d} + W_d \cdot t_{dd} + W_s \cdot t_{sd} \end{aligned} \tag{13}$$

where W_s and W_d indicate the nominal weights of the downwelling energy at the bottom in the FOVs of the shallow and deep channels, respectively. Equation (13) is just a symbolic description of the total time delay and not rigorous calculations; however, they can reflect the contribution of each propagation procedure to range difference.

III. Results and Discussion

A. Variation of Range Difference

Typically measured waveforms collected by Mapper5000 and CZMIL are shown in Fig. 5(a) and (d), respectively. Shallow-channel and deep-channel waveforms differ from each other. The shallow-channel waveforms from either Mapper5000 or CZMIL had two peaks representing the surface and bottom returns and exhibited an exponential decay behavior of the backscattering signal from the water column. The most unique feature was the existence of a wide peak within

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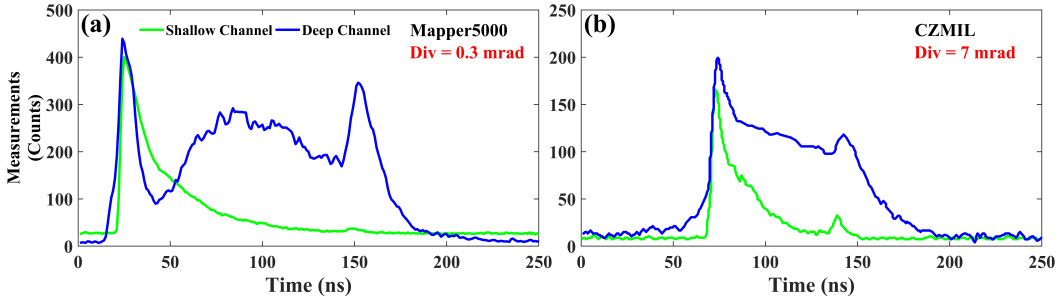

Fig. 5. Measured waveforms of Mapper5000 (a) and CZMIL (b).

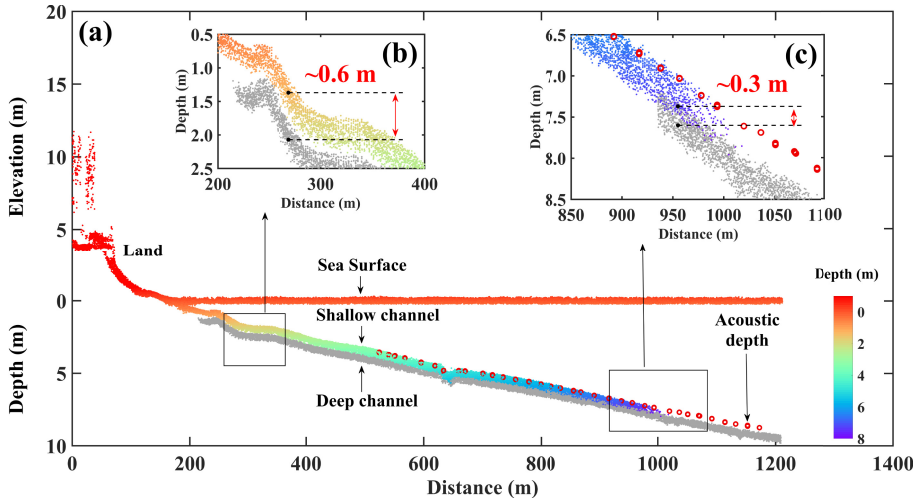

Fig. 6. (a) Profiles of bathymetric topography in the Weihai coast and the acoustic sounding points. The dots show a clear range difference between the shallow and the deep channels. (b) Range difference in the shallow water region is ∼0.6 m. (c) Obvious discontinuity of water depth occurs in the conjunct of two channels, and the range difference is ∼0.3 m.

the water column observed in the deep-channel waveform of Mapper5000 [Fig. 5(a)], which was not observed in CZMIL. These wide peaks can be found in the Mapper5000 waveform datasets at our three study sites, which were not caused by the scattering layers or underwater objects. Such a unique feature mainly originated from the spatial spreading of a laser beam, which was caused by multiscattering of particles. Beam spread adds an additional loss factor, which has been characterized as an increased exponential loss factor because an insufficient FOV spatially excludes a portion of the returning signal. This loss factor reduces the LiDAR backscattering signal in a highly complex way and depends on the FOV of the receiver and the water VSF [50]. Mapper5000 and CZMIL have similar segmented FOV channels; however, the DIV of Mapper5000 is approximately 0.3 mrad and much smaller than 7 mrad of CZMIL. This unique feature in the Mapper5000 waveforms is probably attributed to the DIV difference. More importantly, the beam spreading effect can significantly broaden the bottom returns and influence the determination of range in water. The peak position of the bottom reflection is denoted by the dashed line in Fig. 5. Both Mapper5000 and CZMIL show that the peak position of the deep channel shifted to the right of that of the shallow channel, indicating the deep channel yielded a larger range than the shallow one.

Significant differences in in-depth measurements between the shallow and deep channels were also observed in our experimental results. At the site of the Weihai coast, the depths derived from the deep channel were larger than those from the shallow channel (Fig. 6). In shallow depths near 2 m, the deviation between the two channels is ∼0.6 m [Fig. 6(b)], while it is ∼0.3 m at a deeper depth. The bathymetric results of the two channels show an evident depth discontinuity (approximately 7.5 m depth) at the junction of shallow and deep channel measurements [Fig. 6(c)]. The bathymetric results from the LiDAR, particularly from the deep channel, are greater than the acoustic depth measurements. Thus, the range difference between the two channels shows a large variation with depth and can easily exceed the IHO accuracy standards. Similar observations were found at the two other sites, indicating that the range difference prevailed and must be corrected.

The bottom returns in the shallow-channel and deep-channel waveforms collected by Mapper5000 from the three sites are displayed in Fig. 7(a)–(c). The shallow-channel waveforms are denoted by green lines; and the deep-channel waveforms by blue lines. The peak position of the bottom reflection is denoted by the dashed line. The peak position or range difference between the two channels prevailed in various water bodies. The peak position of the deep channel shifting to the right of that of the shallow channel was observed at all

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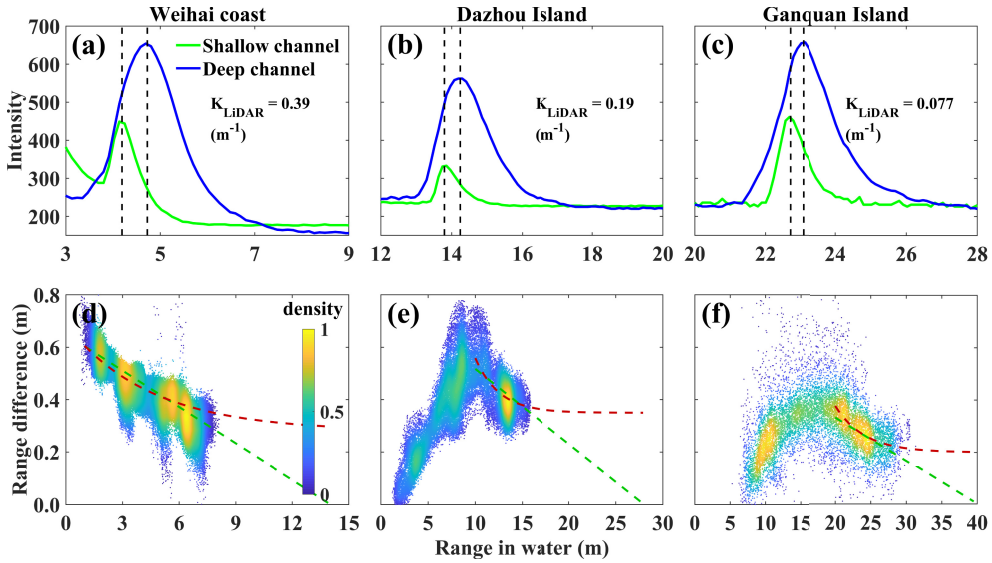

Fig. 7. (a)–(c) Measured bottom return waveforms at three sites, including shallow-channel waveforms (green solid lines), deep-channel waveforms (blue solid lines), peak position (dash lines), and K_{LiDAR}. (d)–(f) Scatter plots of the range difference between shallow and deep channels versus range in water from the shallow channel at the three sites; point density is denoted in different colors. Two lines reveal the potential changing tendency of range difference between the two channels. The green line is a direct decline, while the red line is decreasing gradually to a constant.

sites, indicating that the deep channel yielded a larger range than the shallow one. During the propagation in the water body, the laser pulse suffered from multiscattering effects, and the spread gradually deviated from the forward direction. The transfer path length between the surface and bottom was elongated with the increase in the number of scatterings. The deep channel was more likely to collect multiple scattering light compared with the shallow channel. This notion means that the photons captured by the deep channel had a larger probability to have a longer transfer path, resulting in deeper range measurement.

The scatter plots of range difference against depth are presented in Fig. 7(d)–(f). The bathymetric data in a very shallow depth (<2 m) were discarded because of the overlap between the surface and bottom reflections. The range difference between the two channels changed with water depth. The trend that range difference first increased and then decreased was observed in Dazhou Island and Ganquan Island, although the range difference on the Weihai coast showed a monotonic decrease with depth. Thus, the main feature of these data is the existence of maximum in range difference curves. These maximum values occurred at the inflection point as the range difference trended to switch from increase to decrease. The maximum range difference (MRD) at the three sites ranged from 0.4 to 0.6 m, and the maximum occurred at the depth of ∼10 m on Dazhou Island and ∼20 m on Ganquan Island. K_{LiDAR} was calculated by a linear fit to the logarithm of the water column signal of the shallow channel. K_{LiDAR} for Ganquan Island, Dazhou Island, and the Weihai coast are 0.077, 0.19, and 0.39 m^{-1}, respectively, showing a large difference in the water optical properties of these sites. The amplitude of range difference and variation of maximum position were associated with water optical properties. Thus, the influencing factors of range difference may be complex, and the pulse stretching induced by scattering may play a major role and should be recognized.

Additionally, the maximum detection depth of the shallow channel is significantly smaller than that of the deep channel. The range difference between the two channels at a larger depth where the shallow-channel range data were unavailable cannot be derived from field measurements. As seen in Fig. 7(d)–(f), whether the range difference tendency beyond the maximum depth of shallow-channel data decreased sharply to zero or exponentially to a constant value is not clear. Thus, revealing the range difference tendency across the measured depth range is a challenge and should be accessed further.

B. Range Difference Sensitivities

The observed range difference between two channels by Mapper5000 showed a complicated relationship with depth, and it is quite different from the deviation between lidar and sonar depths, which generally increase with depth [27]. To check whether these complicated range difference trends are caused by instrument faults or by the radiative transfer process of laser in water, and to estimate the range difference between two channels at greater depths, we utilized the SALMOM to generate range difference between the two channels under various optical properties and instrument parameters; we want to confirm these observed relationships of range difference to depth and reveal the radiative transfer mechanism behind them.

1) Water Optical Properties: A series of simulations with various absorption coefficients a, scattering coefficients b, and phase functions were carried out for Mapper5000 to assess the effect of water optical properties. Range differences calculated from these simulated waveforms are presented in Fig. 8. It can

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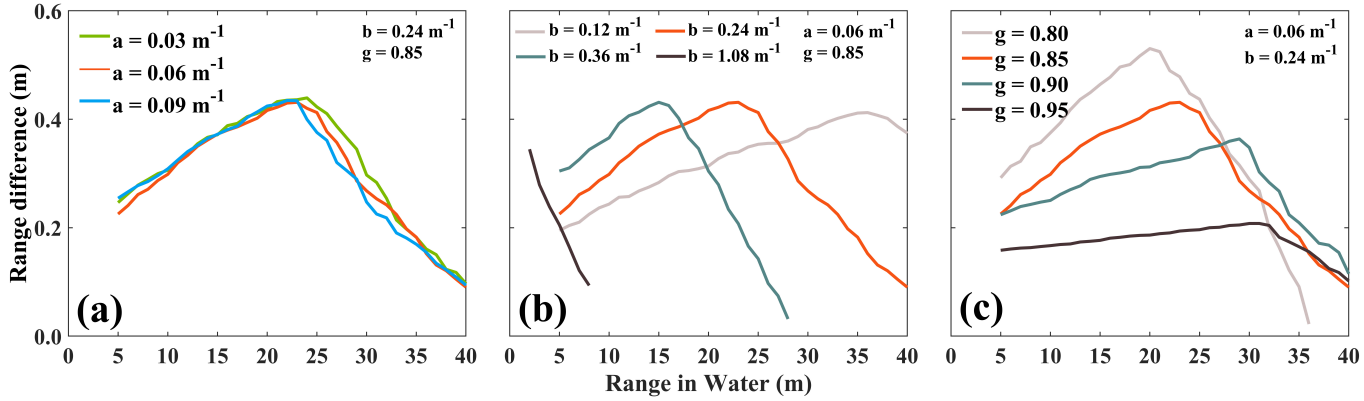

Fig. 8. Influence of different water optical properties on range difference. Relationship between range difference and range in water with different: (a) absorption coefficients; (b) scattering coefficients; and (c) asymmetric factor g of VSF.

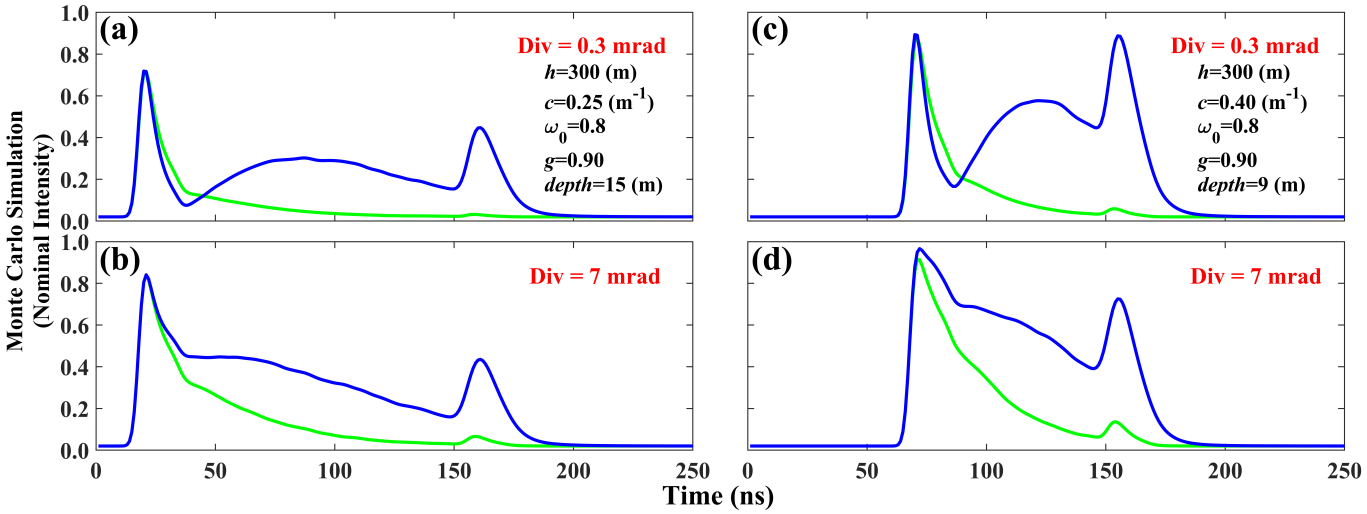

Fig. 9. (a)–(d) Simulated waveforms with different beam divergence angles and different water optical properties.

be found that the trend and magnitude of range difference [Fig. 8(a)] almost did not vary with a (from 0.03 to 0.09 m^{-1}), indicating that the influence from a was limited. By contrast, b significantly affected MRD position, although having a smaller influence on MRD magnitude, when the asymmetric factor of scattering phase function is equal [Fig. 8(b)]. The MRD position shifted to a smaller range with increasing b. This shift can be explained using the similarity theory, wherein the light field characteristics were considered functions of the dimensionless optical depth coordinates when the scattering phase function and ω_0 are equal. Values of ω_0 in these simulations were not identical but comparable; thus, the MRD position almost occurred in the range with similar optical depth. When the scattering coefficient was up to 1.08 m^{-1}, no inflection was observed at the MRD position, similar to the observation on the Weihai coast [Fig. 7(d)], which can be attributed to MRD occurring at a shallow depth. In such a case, since the full-width at half-maximum (FWHM) of the laser pulse is not zero, the surface and bottom returns would be overlapping at a small depth and pulse stretching may be difficult to distinguish. The range difference at small depths cannot be well simulated and determined.

The scattering phase function has a more complicated impact on range difference because it has the closest association with pulse stretching. In Fig. 8(c), g increased when a and b were fixed, indicating that forward scattering was stronger, causing not only the MRD position to shift to a greater depth, but also the MRD to decrease. Beyond the range of the MRD position, the range difference showed a linear decrease with a range in water, which was a useful feature for range difference correction. The laser pulse dispersion causing the range difference showed a higher dependency on the scattering phase function in a complex manner [20], [31], [53]. Therefore, the scattering phase function is the most critical optical feature in determining the range difference. Generally, the results in Fig. 8 coincide with Mapper5000 measurements (Fig. 7) and confirm the dependence of range difference on water depth and optical properties.

2) Beam Divergence Angle: Besides water optical properties, the variation of DIVs also influences LiDAR bathymetry [51]. The difference in DIVs between Mapper5000 and CZMIL impacted the range difference, which showed a significant influence on the waveform of the deep channel. Fig. 9(a)–(d) shows the simulated shallow- and deep-channel

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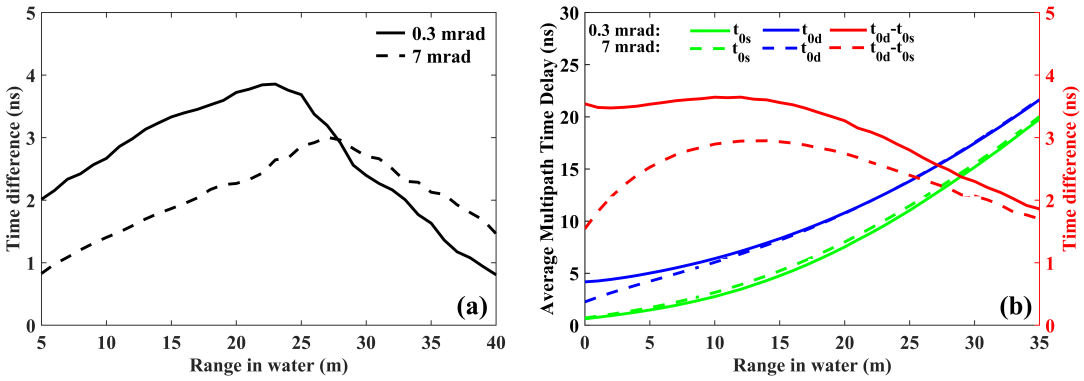

Fig. 10. (a) Range difference between the two channels for different laser beam divergence. (b) AMTD and time difference for the downwelling path.

waveforms for the DIVs of 0.3 and 7 mrad for Mapper5000 and CZMIL, respectively. The SALMOM simulations [Fig. 9(a)] show good agreement with the measurements of Mapper5000 [Fig. 5(a)]; and the distinct wide peak was modeled. The mechanism underlying the pseudo peak in the deep channel (Fig. 9) is as follows: when the laser pulse propagated into the small physical or optical depth, the backscattered energy was almost concentrated in the small FOV. As transferring into deeper depth with increasing multiple scattering, the beam spreading extent was expanded out of the shallow-channel FOV into angular ring FOV, leading to the signal increase of the deep channel. Due to the strong absorption and scattering in water, the signal in the deep channel decreased gradually with depth or time, resulting in a pseudo-broad peak within the water column.

This pseudo-broad peak will not disappear as the attenuation coefficient increases. But the peak position shifts to a shallower depth [Fig. 9(c)]. However, according to Fig. 9(a) and (b), when the DIV was increased to 7 mrad, which was larger than the FOV of the shallow channel (6 mrad), the pseudo peak became indistinct and even disappeared in the deep-channel waveform. When scattering became strong, the laser pulse tended to damp quickly and the backscattering energy from the water column became larger. The attenuation of the signal from the water column in the deep channel was obviously slower than that in the shallow channel, due to the energy supplement from the shallow channel. Basically, the beam spreading and the mismatch between DIV and receiver FOV resulted in this pseudo peak, while water optical properties had significant influences on the magnitude and position of the pseudo peak. More importantly, from the analysis above, the SALMOM method can successfully model the waveforms of segmented FOV channels under a wide range of environments. It further confirms that our previous simulation results and findings are of practical significance.

Second, as presented in Fig. 10(a), the change of DIV from 0.3 to 7 mrad made a decline in the MRD and a slight shift of MRD position to a larger range. Since the segmented FOVs in the simulation were not changed, the influence of DIV variation on range difference was attributed to the AMTDs of the downwelling path (t_{0d} and t_{0s}). Because the 7 mrad DIV was larger than the FOV of the shallow channel, there was energy distribution of incident laser light in the angular FOV of the deep channel once the laser pulse penetrated the water. The time difference for the 7 mrad DIV was smaller than that for 0.3 mrad at all depths [Fig. 10(b)]. Despite variation caused by varying DIV, the tendency that range difference increased first and then decreased linearly to zero remained similar. This further demonstrates that these complicated range difference trends were not only related to the instrument self, but also closely related to the interaction between the laser pulse and water column.

3) Different Propagation Paths: The variation in influences of the scattering phase function on the three paths is revealed, to illustrate the radiative transfer processes and mechanisms behind the range difference occurrence. The results for each propagation path with different scattering phase functions are presented in Fig. 11. The MRD position is represented by the red bar, and the range difference of the three paths is denoted by the time difference calculated by using the AMTD from the two channels and indicated by the red line. The AMTD for each path exponentially increased with range or depth, and the deep channel yielded higher AMTD than the shallow channel in all cases. The AMTD increase was smaller for the higher g case. The incident laser downwelling time difference decreases with an increasing range in water, and the cross-channel upwelling time difference monotonically decreased. Only in-channel upwelling time difference presented a similar trend with the total range difference, which changed from increasing to decreasing with depth. Although subtle deviations can be observed between the positions of in-channel upwelling maximum AMTD and the maximum total range difference, they could be caused by different bottom upwelling energy distributions in the two-channel FOVs.

These phenomena can be explained by the transition between the forward-scattered and diffusion regimes [31]. In the forward-scattering regime with a small physical range or optical thickness, the shallow- and deep-channel receivers captured the forward-scattering signals of the bottom reflection in the FOV of the shallow channel. The total range differences were not associated with the time difference between t_{0s} and t_{0d}, but determined by that between t_{sd} and t_{ss}. Although t_{0d} was approximately 3 ns higher than t_{0s} in the three cases,

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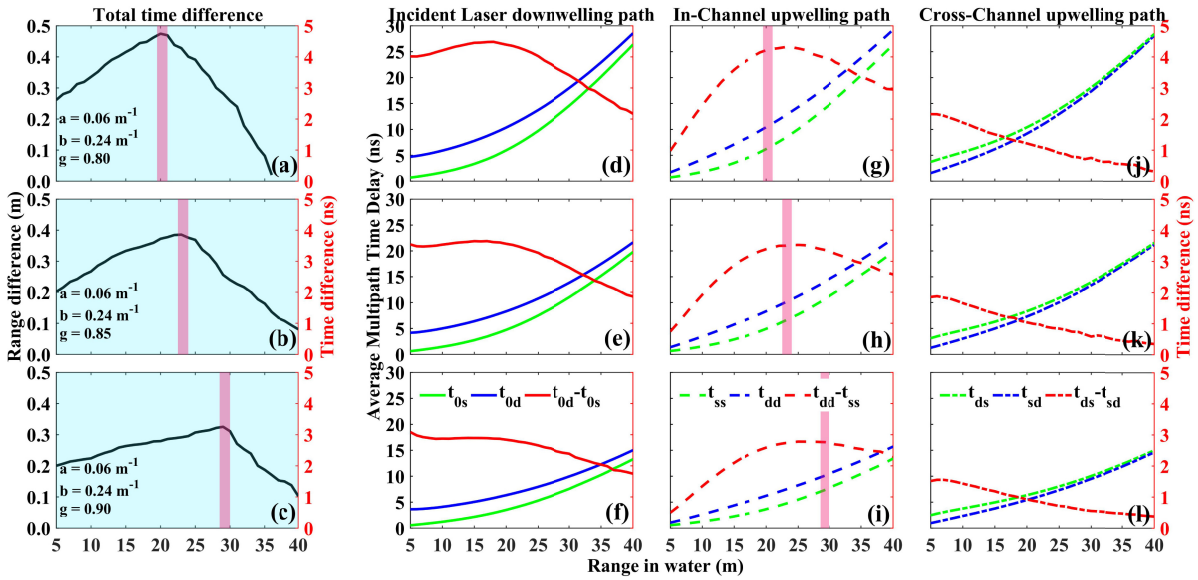

Fig. 11. (a)–(c) Total range difference between the shallow and deep channels for different g values of (a) 0.80, (b) 0.85, and (c) 0.90. a and b are fixed values of 0.06 and 0.24 m^{-1}, respectively. The MRD position is represented by the red bar. (d)–(f) Incident laser average arriving time, (g)–(i) in-channel signal backscattering average arriving time, and (j)–(l) cross-channel signal backscattering average arriving time is displayed from the second to the fourth column, respectively. The time difference between the three paths is displayed by the red line.

the total range differences were lower than 3 ns due to small W_d and the minor deviation between t_{sd} and t_{ss} at small ranges. Scattering out of the forward direction generated more diffusive scattering behaviors with an increasing range of optical thickness, and more downwelling energy was distributed into the bottom region in the deep-channel FOV. Accordingly, most signals received by the shallow and deep channels were from the bottom reflection in the FOV of their own channels, and the total range differences were influenced by the time difference in downwelling ($t_{0d}-t_{0s}$) and in-channel upwelling ($t_{dd} - t_{ss}$) paths. In the case of transition from forward-scattered and diffusion regimes, t_{0d} and t_{dd} of the deep channel were all larger than t_{0s} and t_{ss} because the deep-channel receiver collected more diffused scattering signals; thus, the total range differences increased with depth. When the system approached the diffused regime, the total range differences would achieve the maximum and then decrease because the amount of downwelling diffused energy in the small FOV of the shallow channel was increased via the enhanced temporal dispersion of the laser pulse, and the downwelling time difference ($t_{0d}-t_{0s}$) decreased. Meanwhile, the in-channel upwelling path also showed a decline in the time difference between t_{dd} and t_{ss}. The range differences in all cases would finally disappear because the photons in both channels went through the same diffusion process. The increasing g resulted in stronger forward scattering, causing more scattered photons to remain in the forward direction. Based on the analysis in [31], with increasing g, the photons are more concentrated in the forward scatter region and reach the diffused regime later, and thus the MRD appears at larger depths. Additionally, a larger forward scattering resulted in smaller average propagation times between the bottom and the laser or receiver and reduced the AMTD for each path and the amplitude of the MRD.

C. Range Difference Correction Method

Based on the detailed analysis of measured and simulated results in Sections III-A and III-B, we propose a correction method for the range difference between channels and depth bias in the shallow channel. First, as seen in Fig. 12, the deep-channel depths before range difference correction (denoted in green circles) have much larger depth biases from the sonar depth than the shallow-channel ones. Depths from the shallow channel are closer to sonar depths and thus can be used as a "reference" to correct the range difference between the deep and shallow channels. We use a cubic polynomial function fitting the range difference data to correct the deep-channel data across the whole depth range as follows:

$$z_{\mathrm{cor}} = \sum_{i=0}^{3} k_i \cdot z^i \tag{14}$$

where z is the range in water from the deep channel, z_{cor} is the range difference that can be applied to the deep-channel data as a corrector, and the values of k_i were obtained from the regression of in-water range and range difference data within the maximum detection depth of the shallow channel. The cubic polynomial function showed a good fitting outcome on both measured and simulated datasets. When fitting the simulated results in Fig. 8, all of the residual errors between fitting curves and simulated ones remained within ± 0.03 m. The correction functions for the Weihai coast and Ganquan Island are shown in Fig. 12(a) and (d), respectively. Fig. 12(b) shows the correction results of the Weihai coast. The bathymetric results of the two channels showed good consistency after the correction, and the depth discontinuity at the junction of shallow- and deep-channel measurements of approximately 7.5 m depth was eliminated. Moreover, the deviation between bathymetric results and sonar depth

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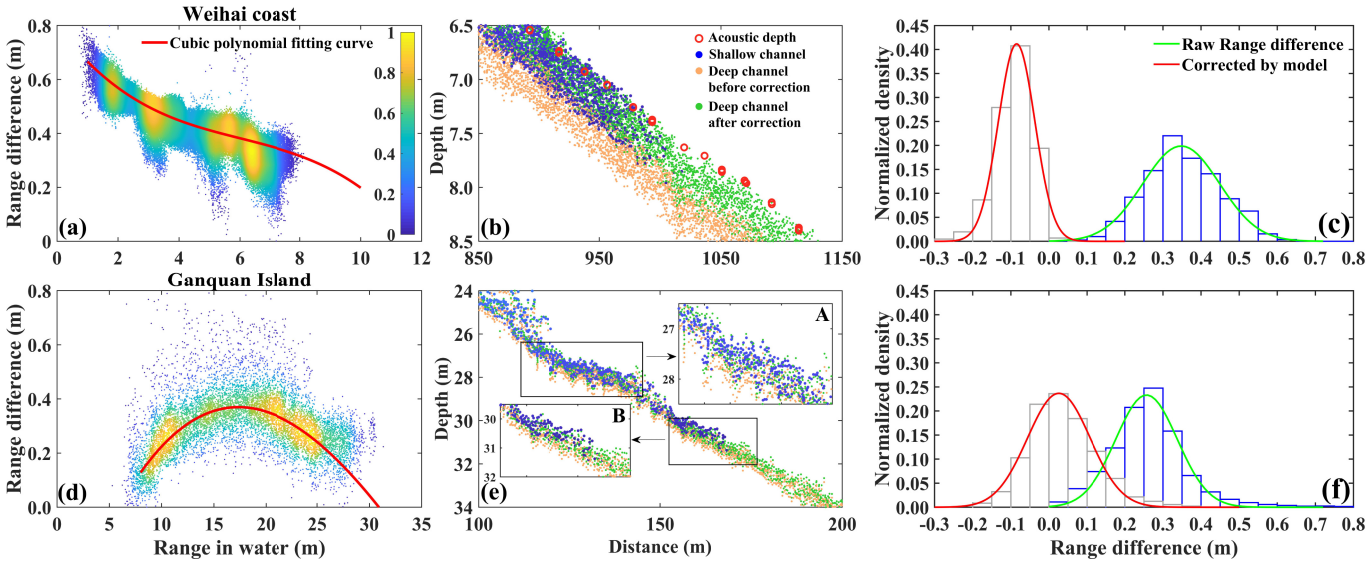

Fig. 12. Scatter plots of range difference against in-water range in (a) Weihai coast and (d) Ganquan Island. The cubic polynomial fitting function for the range difference correction is shown as the red line. (b) and (e) Are bathymetry cloud points after the range difference correction results at the two sites. (c) and (f) Probability density function curves of raw and corrected range difference, which are the green and red solid lines, respectively.

TABLE IV
STATISTIC PARAMETERS OF THE RANGE DIFFERENCE BEFORE AND AFTER CORRECTION

Site	Range Difference (m)	Max. (m)	Mean (m)	Std. (m)
Ganquan Island	Raw	0.792	0.261	0.103
	Corrected	0.702	0.004	0.092
Dazhou Island	Raw	0.826	0.388	0.151
	Corrected	0.517	0.002	0.089
Weihai coast	Raw	0.788	0.353	0.098
	Corrected	0.501	-0.003	0.047

measurements decreased. Similar results were found in the Ganquan Island measurements [Fig. 12(d)]. The probability density functions of the original and corrected range difference are shown in Fig. 12(c) and (f), respectively, while some statistic parameters are displayed in Table IV. The correction method made the range difference in water between channels drop to the same level as that in the air, and the mean range difference and standard deviation were near 0.01 and 0.1 m, respectively. When the range difference among the two channels had been effectively minimized, the corrected Mapper5000 deep-channel depths can also be regarded as shallow-channel data.

Second, the shallow-channel depths still have small biases from the sonar depth measurements, according to the results from the Weihai coast and the Dazhou Island (Fig. 13). The depth biases of the deep channel after correction have the same trend as those of the shallow channel at the two sites. Particularly, the deep-channel data points beyond the maximum detection depth of the shallow channel are evenly distributed around the fitting curve derived from the regression of shallow-channel data and sonar depths, which indicates that the range difference correction for the deep channel is reasonable. As mentioned above, these remained biases are also caused by spatial and temporal dispersion of the laser pulse and definitely are functions of water optical properties, depth, nadir angle, and FOV. Guenther and Thomas [18] calculated the mean bias with various combinations of known parameters for both linear fractional threshold (LFT) and complex Hydrographic Airborne Laser Sounder (HALS) detection algorithms [52] and applied it as the bias corrector, which was then functions of only nadir angle and depth. It should be noted that these mean bias correctors are developed for the full FOV channels, which is another reason for correcting the range difference between shallow and deep channels before applying these correctors to the deep channel with angular FOV. Since the beam incident nadir angle of Mapper5000 is only distributed in the range of 10.5°–15°, such variation of nadir angle has a very slight influence on bias estimation. For convenience, we used the nadir angle of 15° to calculate these biases. These mean bias corrector models are shown in Fig. 13. Comparatively, the model of LFT 50% seems to give the best result, which is in the form of

$$D_{\text{bias}} = 0.065D^{0.58} - 0.27D^{1.25}(1 - \cos(15°))^{1.26} \quad (15)$$

where D is the depth from the deep channel in meters. The mean bias corrector of LFT 50% makes the uncertainty of the final LiDAR depth at the Weihai coast and Dazhou Island within 0.2 m, which satisfies the IHO standards. Thus, the

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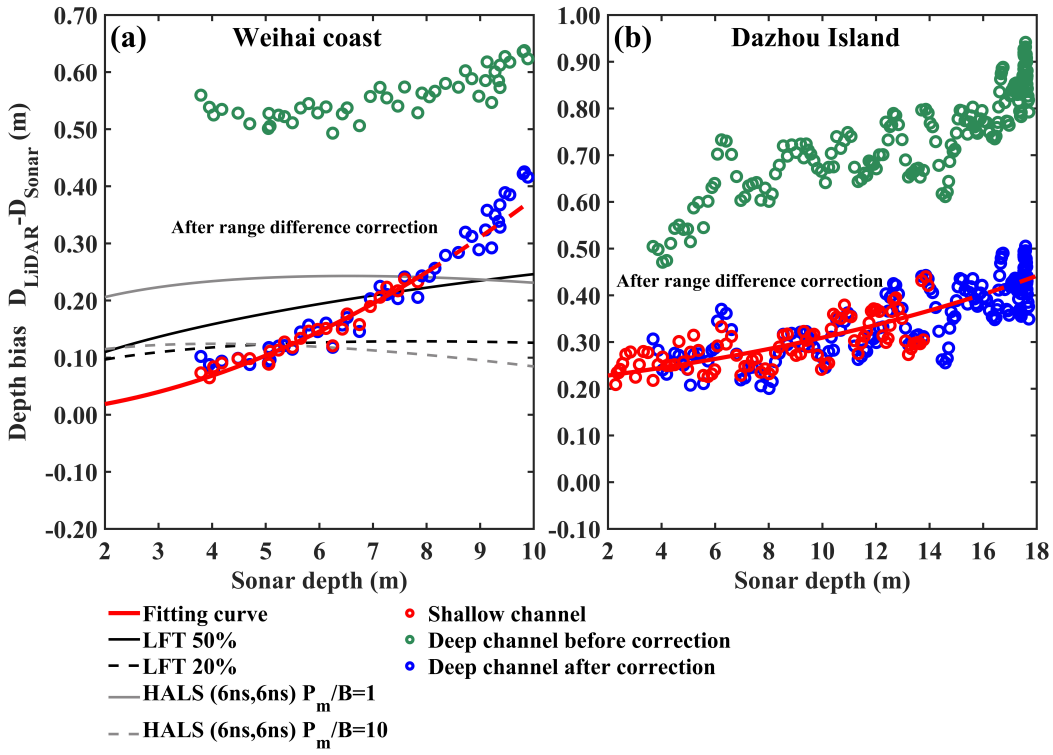

Fig. 13. Depth bias for shallow-channel data (red circles) and deep-channel data (green circles) at (a) Weihai coast and (b) Dazhou Island; and mean bias corrector models (LFT and HALS) from Guenther and Thomas (1984). The depth biases after range difference correction and mean bias corrector of LFT 50% for shallow channel and deep channel (blue circles) are also presented. The red curves are derived from regression of shallow-channel data and sonar depth, showing the relationship of LiDAR depth bias to sonar depth before depth bias correction of LFT 50%.

combination of range difference correction and mean bias corrector can be an alternative method for depth bias correction of segmented FOV bathymetry LiDAR when the referenced sonar data are not available.

IV. Discussion and Conclusion

Significant range differences were identified between the shallow and deep channels of the Mapper5000 bathymetry lidar system with segmented FOV receivers. The experiments at the three sites revealed that the range difference varied with depth and water optical properties. The main feature of these data was the existence of a maximum in the range difference curves, which ranged from 0.3 to 0.6 m and exceeded the IHO accuracy standard. The SALMOM, accounting for the oblique geometric model of the LiDAR system, was developed to assess the effects of water optical properties, FOVs, and DIVs on range difference. The simulation results confirmed the trend in Mapper5000 measurements and indicated that the range difference showed linear decreasing beyond the depth of MRD and finally tended to zero. Furthermore, compared to the optical properties of a and b, the scattering phase function and DIV played more important roles in determining the amplitude and position of the MRD in these curves. The downwelling path, as well as in-channel and cross-channel upwelling paths, were simulated by the Monte Carlo model to investigate the contributions of different propagation paths to range difference. It was indicated that the range differences at different depths were to some extent attributed to different propagation paths, and the underlying physics was the transition of laser propagation between the forward-scattered and diffusion regimes. A range difference correction method by fitting existing shallow- and deep-channel data in the overlapping range with a cubic polynomial was proposed to correct the deep-channel data in the whole in-water range. The depth discontinuity at the junction of shallow- and deep-channel measurements was removed, and the mean and standard deviation of corrected range difference were near 0.01 and 0.1 m, respectively. The depth bias that remained in the shallow-channel data can be further reduced by applying the mean bias corrector of LFT 50%, and the final uncertainty of the LiDAR depth was within 0.2 m at both Weihai coast and Dazhou Island, which satisfies the IHO standard. Thus, a combination of range difference correction and mean bias corrector can be an alternative method for depth bias correction of segmented FOV bathymetry LiDAR when the referenced sonar data are not available.

However, there are still limits in the current bias corrector for the shallow channel (Fig. 13), since the mean bias corrector seemed insufficient due to the remaining variation from these means to the bounding values [18]. It is indicated that these mean bias corrector models may not be suitable for all water bodies with different optical properties. In fact, the prediction of remained bias of shallow channel is difficult, since some independent parameters, particularly the optical parameter of b and scattering phase function, are unknown. Although many researchers have conducted studies to obtain turbidity from

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waveforms [53]–[55], b and scattering phase functions are still difficult to estimate in real-time from LiDAR returns. Any future improvement should involve more precise depth bias correction for shallow-channel depth, such as the utilization of segmented FOV LiDAR waveforms of themselves to retrieve the b and scattering phase function and then predict the depth bias.

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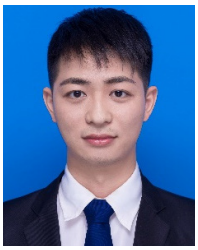

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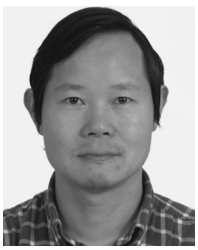

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